arXiv:2410.16090v2 [cs.CL] 9 Feb 2025

Analysing the Residual Stream of Language Models Under Knowledge Conflicts

Yu Zhao[1] **Xiaotang Du**[1] **Giwon Hong**[1] **Aryo Pradipta Gema**[1] **Alessio Devoto**[3]
Hongru Wang[2] **Xuanli He**[4] **Kam-Fai Wong**[2] **Pasquale Minervini**[1,5]
[1]University of Edinburgh [2]The Chinese University of Hong Kong
[3]Sapienza University of Rome [4]University College London [5]Miniml.AI
{yu.zhao, p.minervini}@ed.ac.uk

Abstract

Large language models (LLMs) can store a significant amount of factual knowledge in their parameters. However, their parametric knowledge may conflict with the information provided in the context. Such conflicts can lead to undesirable model behaviour, such as reliance on outdated or incorrect information. In this work, we investigate whether LLMs can identify knowledge conflicts and whether it is possible to know which source of knowledge the model will rely on by analysing the residual stream of the LLM. Through probing tasks, we find that LLMs can internally register the signal of knowledge conflict in the residual stream, which can be accurately detected by probing the intermediate model activations. This allows us to detect conflicts within the residual stream before generating the answers without modifying the input or model parameters. Moreover, we find that the residual stream shows significantly different patterns when the model relies on contextual knowledge versus parametric knowledge to resolve conflicts. This pattern can be employed to estimate the behaviour of LLMs when conflict happens and prevent unexpected answers before producing the answers. Our analysis offers insights into how LLMs internally manage knowledge conflicts and provides a foundation for developing methods to control the knowledge selection processes.

1 Introduction

Large language models (LLMs) have shown remarkable capability to memorise factual knowledge and solve knowledge-intensive tasks [Petroni et al., 2019, Brown, 2020, Touvron et al., 2023, Jiang et al., 2023, Team et al., 2023]. Nevertheless, the knowledge stored in their parameters (*parametric knowledge*) can be inaccurate or outdated. To alleviate this issue, retrieval and tool-augmented approaches have been widely adopted to provide LLMs with external knowledge (*contextual knowledge*) [Karpukhin et al., 2020, Lewis et al., 2020, Wu et al., 2022, Schick et al., 2024]. However, such contextual knowledge can include information that conflicts with the parametric knowledge of the model, which may result in undesired behaviour; for example, the model can rely on inaccurate information sources and produce inaccurate generations [Mallen et al., 2023, Xie et al., 2024, Su et al., 2024, Wang et al., 2023, Hong et al., 2024, Zhao et al., 2024a].

Prior research found that LLMs tend to prefer contextual knowledge (e.g. retrieved passages) over their parametric knowledge [Su et al., 2024, Xie et al., 2024]. However, in more general applications, LLMs should retain the ability to use parametric knowledge when presented with incorrect or undesirable information [Chen and Shu, 2023b,a, Zou et al., 2024, Mallen et al., 2023, Zhong et al., 2023]. To achieve this goal, LLMs are expected to acknowledge the existence of conflicts, allowing

Accepted at the Foundation Model Interventions Workshop @ NeurIPS 2024. This work is the preliminary study of "Steering Knowledge Selection Behaviours in LLMs via SAE-Based Representation Engineering"

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

them to alert the user while keeping the decision-making process under the user's control for further action. Existing works investigate the fine-tuning and prompting-based strategies to detect knowledge conflicts [Wang et al., 2023]. These methods need additional interactions with the model, e.g., by asking the LLMs to examine the conflicts sentence by sentence [Wang et al., 2023], which may result in high latency times and prevent practical applications of these models. Additionally, they do not provide insight into how LLMs internally detect and resolve conflicts.

In this work, we analyse the residual stream [Elhage et al., 2021, Olsson et al., 2022] in LLMs to better understand their behaviour when knowledge conflicts arise, especially between parametric knowledge and contextual knowledge. Our probing experiments on the residual stream indicate that the signal of knowledge conflict rises from the intermediate layers (e.g., the 13th layer of Llama3-8B). Utilising this signal, a simple logistic regression model can achieve 90% accuracy in knowledge conflict detection without modifying the input and parameters of LLMs while introducing only a negligible computation overhead. Moreover, we also observe that the residual stream exhibits different patterns starting from the middle layers (e.g., the 17th layers of Llama3-8B) when the model takes different source information to resolve the conflict. For example, when the model uses contextual knowledge, the residual stream exhibits a significantly more skewed distribution compared with when it uses its parametric knowledge.

In conclusion, our analysis of the residual stream reveals that: 1) LLMs exhibit internal mechanisms for identifying conflicts, and this signal can be leveraged to detect conflicts effectively in the mid-layers of LLMs; 2) LLMs display distinct skewness patterns in the residual stream when using different sources of information, which provides insights on the model's behaviour.

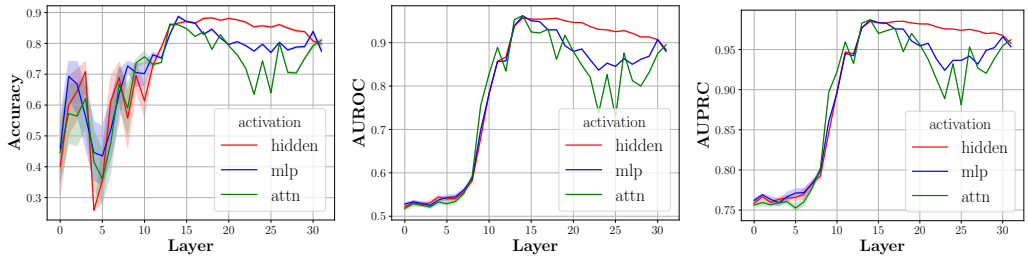
2 Background and Methods

Residual Stream We examine the Transformer architecture from the perspective of the residual stream [Elhage et al., 2021, Olsson et al., 2022]. In this framework, tokens flow through the model, with their embeddings being modified by vector additions from the attention and feed-forward blocks in each layer. We denote the hidden states at position i at l-th layer as $\mathbf{h}_i^l \in \mathbb{R}^d$, where d is the dimension of the internal states of the model. The model produces the initial residual stream $\mathbf{h}_i^0$ by applying an embedding matrix to the tokens. Then, the model modifies the residual stream by a sequence of L layers Transformers, where each Transformer layer consists of a Self-Attention block and MLP at l-th layer. Formally, denote $\mathbf{a}_i^l$ and $\mathbf{m}_i^l$ as the activations of Self-Attention and MLP respectively, the update of the residual stream at l-th layer is $\mathbf{h}^{l'} = \text{LayerNorm}(\mathbf{h}^{l-1}) + \mathbf{a}_i^l$ and $\mathbf{h}^l = \text{LayerNorm}(\mathbf{h}^{l'}) + \mathbf{m}_i^l$.

Linear Probing Linear probing [Conneau et al., 2018, Zhu and Li, 2023, Allen-Zhu and Li, 2023] is a commonly used technique to analyse whether certain information is encoded within the residual stream of a language model. Specifically, for an activation $\mathbf{x}$ from the residual stream, i.e., $\mathbf{h}$, $\mathbf{a}$, or $\mathbf{m}$, a logistic regression model is applied to perform binary classification: $P(y = 1|\mathbf{x}) = \delta(\mathbf{x}\mathbf{W})$, where $\mathbf{W} \in \mathbb{R}^{d\times 1}$ is the learned weight that linearly projects the activation into a scalar value , and δ is the Sigmoid function that outputs the likelihood of probed information existing in the activation.

3 Experimental Setup

Problem Setup Following previous studies [Longpre et al., 2021, Hong et al., 2024, Xie et al., 2024, Su et al., 2024, Wang et al., 2023], we use open-domain question-answering (ODQA) tasks to investigate the behaviours of LLMs when there is a conflict between the model's parametric knowledge and contextual knowledge. In ODQA datasets with knowledge conflicts, each instance is presented as (q, e_M, e_C, a_M, a_C), where q is the question, e_M is the evidence that supports the memorised knowledge, e_C is the evidence that conflicts with the language model's memorised knowledge, a_M is the answer based on and e_M, and a_C is the answer based on the e_C. The model's parametric knowledge is tested in the close-book setting, where the model generates answer a_M based on the question q without external evidence. We generate the answers using a greedy decoding strategy. We use three in-context demonstrations to align the answer format and, for fairness, use the same in-context demonstrations in all experiments.



(a) Accuracy (b) AUROC (c) AUPRC

Figure 1: Accuracy, AUROC, and AUPRC of probing models on detecting the knowledge conflicts based on the activations of Llama3-8B. The probing results on hidden state, MLP and Self-Attention activation are coloured red, blue and green, respectively. More analysis is presented in Appendix B.

Datasets and Models We use NQSwap [Longpre et al., 2021], Macnoise [Hong et al., 2024] and ConflictQA [Xie et al., 2024] to analyse the residual stream when knowledge conflicts arise. We present the experiment results of NQSwap using Llama3-8B [Dubey et al., 2024] in the main paper, and the results of other datasets and models are provided in Appendix B and Appendix C. The training details of the probing model are presented in Appendix A

4 Results and Findings

In this work, we aim to answer the two following research questions: *1)* Can we identify the conflict between context and parameter knowledge by probing the residual stream? *2)* Can we know which source of knowledge the models will use before they generate the answers? We probe and analyse the residual stream to answer these two questions in the following parts.

Identifying Knowledge Conflicts by Probing the Residual Stream We analyse whether language models can identify contextual-parametric knowledge conflicts by probing the residual stream. To this end, we create two groups of instances, $D^{e_C} = \{(q, e_C)\}$ and $D^{e_M} = \{(q, e_M)\}$, where the model generates answers based on conflict evidence in D^{e_C} and non-conflict evidence in D^{e_M}. The probing model is trained to classify whether a given activation is from D^{e_C} or D^{e_M}. We probe the residual stream at the final position to determine if the model is aware of the conflict during the first token generation. This is because the hidden state at the last position in the output layer is used to predict the first token of the answer. For each activation $\mathbf{h}^l$, $\mathbf{a}^l$ and $\mathbf{m}^l$ at each layer, we train a probing model to classify whether it belongs to D^{e_M} or D^{e_C}.

As shown in Figure 1(b) and Figure 1(c), the AUROC and AUPRC of the probing models increase from the first layer to the 14th layer, and this trend is same across the hidden state, MLP, and Self-Attention activations. In Figure 1(a), the accuracy of the probing models at the early layers is random; similar to the trend of AUROC and AUPRC, the accuracy also reaches the highest score at the 14th layer. The above observation indicates that the residual stream does not contain information about knowledge conflict at the early layers. This information rises from around the 8th layer and reaches the highest point at the 14th layer.

After the 14th layer, the probing model's performance decreases slightly until the last layer. Besides, we also observe that the probing results of MLP and Self-Attention activations show a significantly lower accuracy than the hidden state after the 14th layer, which may suggest that MLP and Self-Attention do not provide further conflicting information into the residual stream. We find the same trend using Llama2-7B as shown in Figure 4.

Analysis of the Residual Stream When LLMs Using Different Sources of Knowledge We investigate the distribution patterns of the residual stream when the language model uses different sources of information to generate the answer. Based on the model's predictions on instances belongs to D^{e_C}, we classify them into two groups: $D^{e_C}_{a_C}$ and $D^{e_C}_{a_M}$. Here, $D^{e_C}_{a_C}$ represents the set of instances where the model's predictions align with a_C, while $D^{e_C}_{a_M}$ contains the instances where the predictions